



assignment 1 simulink

Linear Control Systems

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Assignment 7: Simulink Implementation and Results

In this part, we reproduced the nonlinear spring–mass–damper model inside Simulink. We used a MATLAB Function block to implement the two state equations, written as

```
function dxdt = f(t, x)

% excitation
w = 100;           % changed manually to 1, 10, 100 for each run
u = sin(w*t);

% nonlinear state equations
dx1 = x(2);
dx2 = -0.12*x(2) - 400*x(1) ...
      - 10000*(x(1))^3 + 10*u;

dxdt = [dx1; dx2];
end
```

We connected the outputs of the MATLAB Function block to two Integrator blocks so that the first integrator produced the displacement $x_1(t)$ and the second produced the velocity $x_2(t)$. A Clock block supplied the time variable, and a Scope block recorded the displacement. This Simulink structure corresponds to the same state-space model we previously solved using ode45, so the goal here was simply to replicate the numerical integration in a block-diagram environment.

We simulated the response for three excitation frequencies: $\omega = 1$, $\omega = 10$, and $\omega = 100$. The figures below show the resulting displacement signals.

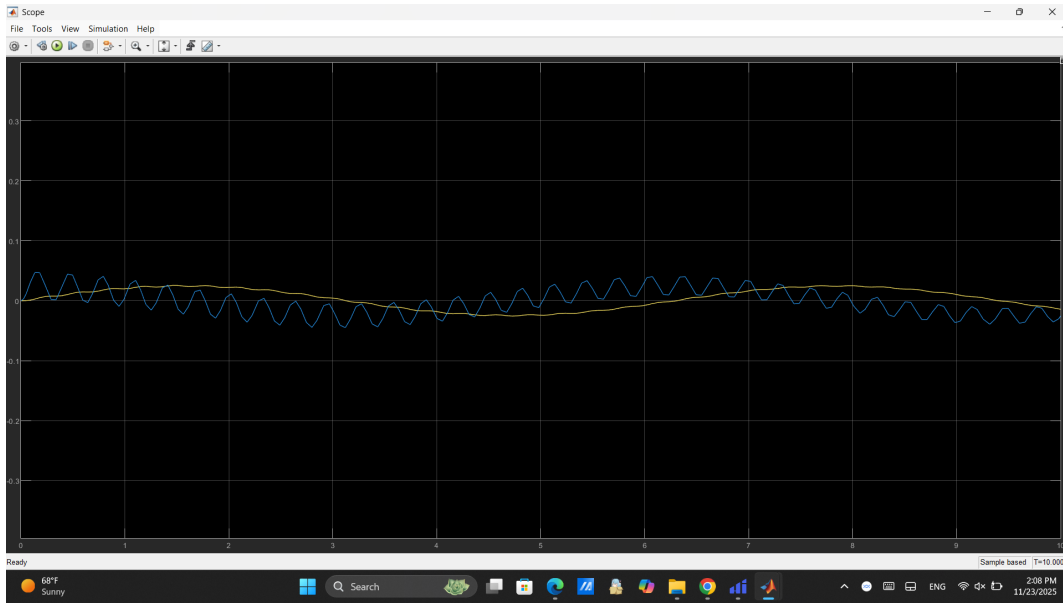


Figure 1: Response for $\omega = 1$. At this low frequency, the motion remained smooth and nearly sinusoidal. The amplitude was small, meaning the nonlinear stiffness term contributed very little, and the system behaved almost linearly.

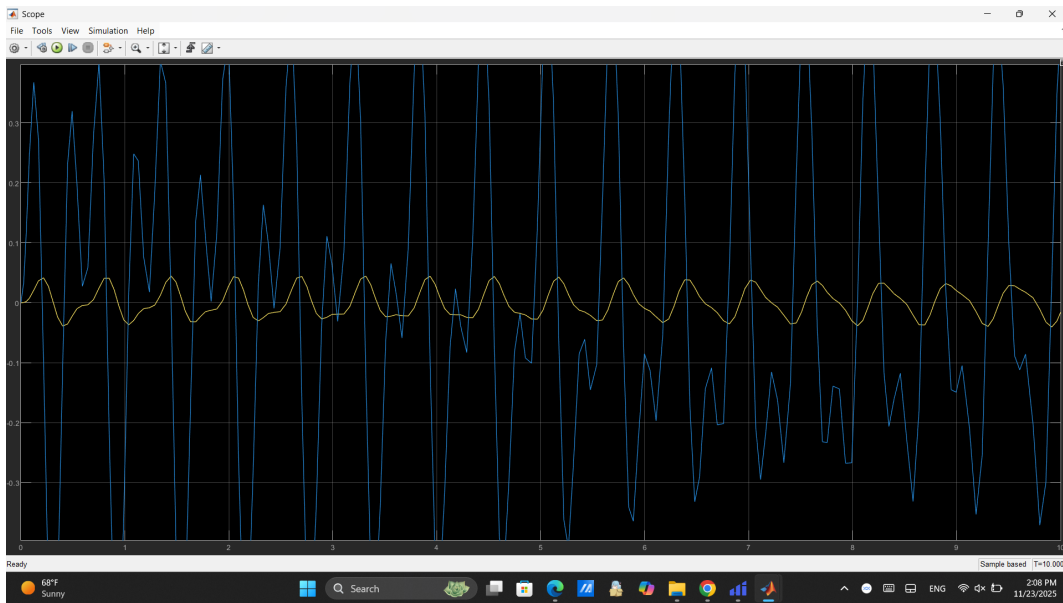


Figure 2: Response for $\omega = 10$. With the excitation closer to the natural frequency, the amplitude increased and the waveform became slightly distorted. This distortion comes from the cubic stiffness term becoming more significant.

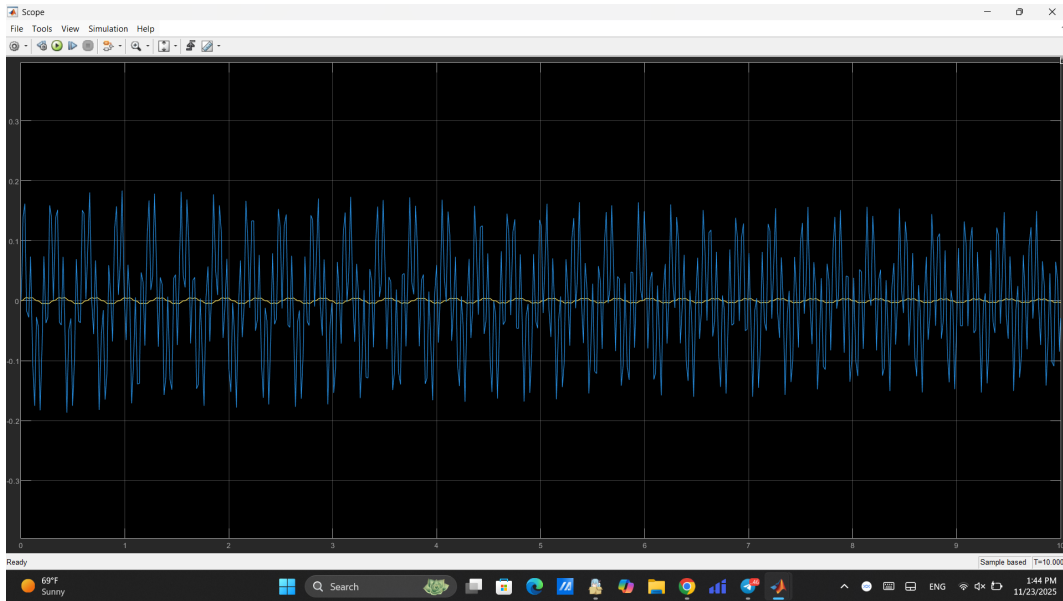


Figure 3: Response for $\omega = 100$. At high frequency, the mass could not follow the rapid input, producing a very small displacement amplitude. The nonlinear effects were again minimal because the motion stayed small.

Overall, the Simulink outputs for the three frequencies matched the behavior we observed when solving the system using `ode45` earlier: low and high frequencies lead to small amplitudes, while mid-range frequencies produce larger and more nonlinear responses.

Assignment 9: Simulink Simulation of the Linear Control System

For this assignment, we recreated the closed-loop antenna control system using Simulink blocks. Earlier, in Assignment 8, we simulated exactly the same linear system in MATLAB using a transfer-function or state-space model with the `lsim` command. Here, our goal was to verify that a Simulink model built from the same LTI dynamics produces identical results.

We ran two simulations: one with a unit-step input and one with a sinusoidal input.

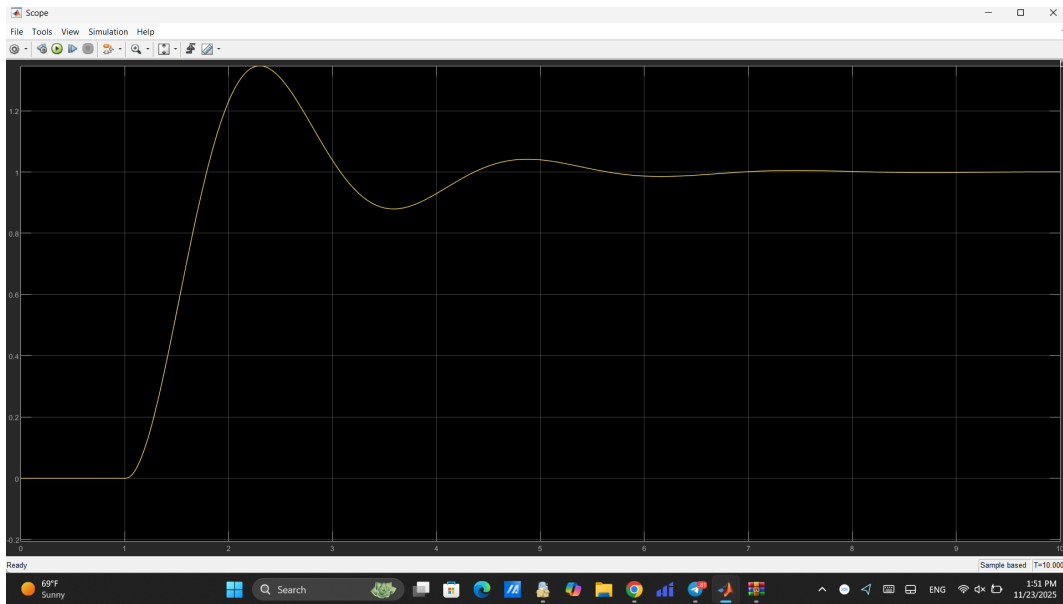


Figure 4: Closed-loop step response. The system rose smoothly, exhibited a small overshoot, and settled at the final value. This matched the MATLAB `lsim` result, confirming that our Simulink diagram accurately represented the same closed-loop model.

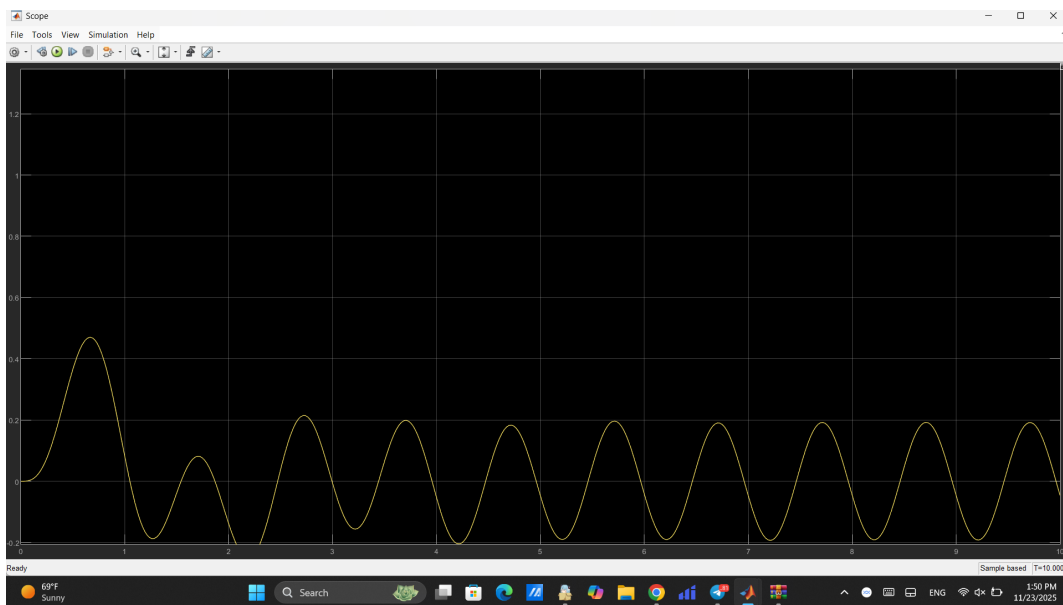


Figure 5: Closed-loop sinusoidal tracking. The system followed low-frequency inputs closely with small phase lag. As the input frequency increased, the tracking accuracy decreased, consistent with the expected bandwidth limits of the closed-loop system. The agreement with our MATLAB simulations verified that both approaches produce the same linear response.

In summary, Assignment 7 allowed us to reproduce our nonlinear `ode45` simulations using Simulink, and Assignment 9 allowed us to reproduce our linear `lsim` simulations in the same way. Together, these two assignments showed that MATLAB and Simulink generate consistent results when the underlying state-space or transfer-function models are equivalent.